Exact Minimax Certificates for Subexponential SAT Contraction

A new finite optimization theorem, sharper contraction bounds, and a review of twenty-nine research packets

Research prepared for Vladimir Reshetnikov

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Abstract

We review twenty-nine refinement packets in the two requested ProveIt paths, including the nine latest arrivals and their subsequent placement, and incorporate the supplied public Anthropic formalization. A new exact finite minimax certificate, explicit four-plan contraction menus, rectangular matrix multiplication and a fully paid recursion schedule improve the best reviewed generic cached contraction bound.

For the shared normalized-wire interface and $E(s) = s/\sqrt{\lceil \log_2 s \rceil}$, the classical-backend construction has time $\mathrm{poly}(s+\mathcal{L})2^{(30.087205604+o(1))E(s)}$ and space $\mathrm{poly}(s+\mathcal{L})2^{(28.852899415+o(1))E(s)}$. Here $\mathcal{L}$ is the input-description length. Adding the published 2026 characteristic-zero matrix bounds gives coefficients 28.972941604 and 28.972651877, respectively. These improve the reviewed displayed cached pair 30.891018921/30.753912833 under the same inherited interface. The gains also survive explicit comparisons that upgrade the older three-plan menus with the same matrix backends and a retuned tail.

The mathematical tool characterizes the exact worst-case support cost of a finite strategy menu as the maximum, over one support chosen from each plan, of the distance from zero to their convex hull. Its rational certificates cover 173,846,817 support tuples. A separately written enumerator reproduces the certificates. Exact rational interval arithmetic checks all schedule gates and the resulting constants.

The logarithmic-order breakthrough is not established. The new algorithms still have exponent of order $s/\sqrt{\log s}$, use subexponential memory, and retain the general-density SAT order $n \log\log n/\sqrt{\log n}$. We give a separate, explicitly conditional route through the actual ranks of Anthropic's completed crossing pieces. A concrete rank-product target would improve the denominator power to $\log_2(3/2) > 1/2$, but that rank theorem remains open. The archive includes the proofs, exact certificates, independently written checks, reproducible source pins and this report's complete LaTeX sources.

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1 Review cutoff and the actual comparison frontier

The mathematical review began at ProveIt commit `73dfb5e5ab6e91bd7375e5b97ef85d7e1cc8e81d`: twenty placed refinement packets and nine archives in `docs/incoming`. The closing review includes commit `7de9d09fecd17de776d712835ecb2011205c531d`. In the intervening two commits, the nine archives were filed under the project and the supplied Anthropic formalization was linked. There are now twenty-nine placed refinement packets; `docs/incoming` contains only its README.

We compared all 443 newly placed files to their archive members using Git blob hashes: all were byte-identical. The placement omits 85 members (the commit describes 84 duplicate source/evidence copies and one checksum ledger). No changed mathematical source was found among the placed files. The closing README now reports the same best bounds independently found in our archive review: 46.18485 in polynomial space and 30.89102 cached. The source and archive hashes, closing commit metadata and per-file relocation check are recorded in the provenance ledger. We also inspect the user-supplied Anthropic commit `1a6eee293b9e00cb432eef6018a21f523623a9af` in pull request 31. No remote repository changes were made.

1.1 Relevant recent commits

Commit	Relevant change
`2b925d8800`	Creates the consolidated SAT project and places seven first-round packets.
`3e09608991`	Vendors the OpenAI preprint, pinned at `7ef5c0cc5c`.
`45233238ae`	Records the initial distinction between density improvements, exponent constants, and unproved stronger-order proposals.
`6d3adc4506`	Places eight second-round packets, including asymmetric packing and the full-Cauchy mode construction.
`450252e9d8`	Places five third-round packets; the displayed generic polynomial-space coefficient reaches 55.70948, cached time 35.35459.
`b9941a7396`	Adds the nine newly reviewed archives. The actual displayed frontier becomes 46.18485 in polynomial space and 30.89102 cached.
`73dfb5e5ab`	Initial review snapshot containing both requested paths and all nine arrivals.
`a7ae5ab55f`	Places the same nine archives under Research and updates the frontier table.
`7de9d09fec`	Closing snapshot: links the published Anthropic Lean development.

The closing review date is October 27, 2026 UTC. Repository author dates use Pacific time and show October 26 for these same commits. All claims are scoped to the pinned revisions, rather than later changes.

1.2 Inventory of the twenty placed packets

"PS" below means polynomial space. Numerical coefficients refer to the generic normalized-wire interface unless the row explicitly states a stronger degree premise. The table reports the packets' claims, with their imported dependencies retained; it is not a certification of every proof in every packet.

Packet	Relevant strongest contribution
`cauchy_mode_improvement`	Factorial Cauchy weights with slope three; the logarithmic order is unchanged.
`eth_contraction_improvements`	Sharper Hermite counts, streamed extraction, and PS coefficient 648.44.
`3sat_precision_refinement`	Leading precision $6MD/u$ and taper minimax; an $s/\log s$ summation interface remains unproved.
`sat_density_improvement`	Density-sensitive envelope (2.5); an actual order gain at intermediate density.
`eth_density_refinement`	Independent density-sensitive improvement and precision savings.
`3sat_refinement`	Scalar parity lift; a compact second-contraction factorization is missing.
`3sat_asymptotic_analysis`	Restricted precision, degree, and rank-encoding barriers; no smaller unrestricted order.
`eth_asymmetric_packet_balancing`	Nearly lossless asymmetric packing; generic PS 76.909.

Packet	Relevant strongest contribution
`3sat_round2_refinement`	Generic PS 84.066, actual-domain rank audit, and review of the alternate manuscript.
`eth_balanced_packing`	Balanced whole-vertex labels, thin precision guard; PS 94.748.
`eth_high_moment`	Independent PS 94.748; stronger low-degree bounds.
`eth_matching_refinement`	Another PS 94.748 derivation using a matching potential.
`eth_packet_refinement`	Pair-load and guard improvements; PS 95.664.
`eth_recursive_refinement`	Refined residue-class modes; PS 133.207.
`eth_joint_cauchy`	Full-block slope-four modes; its ingredients are stronger when composed with later packing.
`sat_adaptive_contraction`	PS 55.70948; cached 36.75031/34.02418; additional low-degree branches.
`sat_frontier_improvements`	Variable arity; PS 63.857, cached 38.484/34.275.
`subexponential_sat_frontier`	PS 63.852, cached 37.930/33.623; exact labeling and precision-peeling kernels.
`subsat_contraction_tree`	Paid recursive group tables and an exact ring bridge to fast matrix multiplication; cached 35.35459/34.27480.
`subexponential_sat_kernel_improvements`	Actual-domain Smith separation, coefficient-extraction kernels, and compatible coefficient compositions.

1.3 The nine new archives

The names below omit only the date suffix and extension. These archives were safely extracted and their report sources, relevant proofs, status files and supplied computational evidence were inspected as source data. Their old benchmark receipts are not measurements made in this session.

Incoming packet	Claimed generic coefficients and substantive change
`3sat_round3_frontier`	PS 63.75736; cached 37.93011/33.62304; useful specialized degree and kernel refinements.
`ETH_Adaptive_Exact_Kernels`	PS 55.33162; cached 35.07061/33.99950; sparse scoring and fused Cauchy transitions.
`eth_adaptive_schedules`	PS 63.78117; cached 37.59170/33.62304; exact incremental packing.
`sat_adaptive_bounds`	**PS** 46.18485, from complete binary linear flags; cached 34.32528/34.02418.
`sat_adaptive_frontier`	PS 50.30469; joint cached 31.25240 time and space; conserved degree-flow identity, with the entry-evaluation gap retained.
`sat_adaptive_matrix_partition`	PS 50.21043; cached 34.32528/34.02418; adaptive partition and separator-tree selection.
`sat_adaptive_tree_certificate`	**Cached** 30.89102/30.75391, from three-plan support averaging and a paid $9 \to 13 \to 7$ schedule.

Incoming packet	Claimed generic coefficients and substantive change
`subexponential_sat_adaptive_four`	PS 50.39373; cached 32.91581/31.66171, or 32.85635/31.46748 with its optional backend; a sharp four-block inequality.
`subsat_rectangular`	PS 53.97753; cached 33.99389/33.99372; genuine rectangular multiplication and a stronger asymmetric-Hermite barrier.

Thus the comparison must use 46.18485 and 30.89102, not the earlier 55.70948 and 35.35459. The new cached methods in this report are additional branches of a portfolio. The linear-flag PS algorithm and every stronger-assumption low-degree method remain applicable.

1.4 Orders, constants, and unresolved targets

No new incoming packet supplies a proved smaller unrestricted logarithmic order. At fixed density the common order is $n/\sqrt{\log n}$; in the worst case it is $n \log\log n/\sqrt{\log n}$. At $m = \Theta(n \log\log n)$, the density-sensitive construction improves the numerator to $n \log\log\log n$. This genuine order improvement was already present in the first round and is not a new result here.

The strongest smaller-order targets still require an unproved algorithmic interface, notably the structured summation in `3sat_precision_refinement`. A numerical coefficient cannot dominate an unproved $n/\log n$ target as though it were an implemented algorithm. Likewise, the independent Anthropic theorem has an unspecified big-O constant in its machine model. It is not numerically comparable with a coefficient per normalized Cauchy wire. Section 8 gives the updated public formalization status.

2 The shared contraction interface and what is imported

Write

$$\lambda = \lceil \log_2 s \rceil, \qquad E(s) = s/\sqrt{\lambda}, \qquad \kappa = \pi\sqrt{2}/\ln 2.$$

The comparison problem is a Boolean tensor network with at most s wires, maximum initial degree at most $\lceil s^{9/10} \rceil$, and initial precision and coefficient truncation length at most s. Its entries lie in the power-of-two coefficient rings of the OpenAI construction. Original entry routines have the prescribed polynomial-description and $2^{O(s^{9/10})}$-time interface. Retained tensors are evaluated at their immutable creation precision and then lifted as specified by the source. These premises matter: an arbitrary exponentially expensive node oracle would not satisfy the theorem.

We import the source's exact deformation, recovery identity, child-network construction, SAT reductions, and their correctness [2, 3]. We also import nearly lossless asymmetric packet packing, the sharp tapered precision, and full-block Cauchy modes from the earlier refinements [5, 4, 6, 7, 8]. This report proves a new method for evaluating the resulting finite mode graph. It does not replace the proofs of the imported identities.

2.1 The exact mode graph

At an active call, let M be its number of wires, u its retained valuation range, and r its fixed alphabet size. After choosing an exact pair-weight allocation, the local answer has the form

$$\sum_{\xi_e \in \mathcal{I}_e} \prod_{g=1}^{r} Z_g\big((\xi_e)_{e \ni g}\big). \tag{2.1}$$

Pairs $e = (g, h)$ are directed and have distinct endpoints. The list $\mathcal{I}_e$ contains the feasible mode-index tuples of the allocated exact weight; Z_g is an actual child-network contraction. Each child has at most $(2/r + o(1))M$ Boolean wires. Empty lists make the allocation zero.

Because r is fixed and all budgets are polynomial in s, the number of exact pair-weight allocations is polynomial. The usual integer knapsack recurrence computes each list size and supplies ranking and unranking with polynomial overhead. This is not a unit-cost counting oracle.

Put $h_e = \log_2 |\mathcal{I}_e|$ and $\mathcal{K}(M, u) = \kappa M/\sqrt{u}$. The imported common-tilt estimate is

$$\sum_e z_e h_e \le \left(\frac{\|z\|_2}{r} + o(1)\right) \mathcal{K}(M, u) \qquad (z_e \ge 0 \text{ fixed}). \tag{2.2}$$

The error is uniform over the allocations and all active calls of the admissible schedules below, for each fixed finite family of z.

For clarity, its leading constants can be recovered directly. The full-Cauchy weights may be bounded below by $w_b = 1 + 4b - 2\,\mathrm{popcount}(b)$. Their logarithmic generating function is

$$\sum_{b \ge 0} -\ln(1 - e^{-\theta w_b}) = \frac{\pi^2}{24\theta} + O\big(\log^2(2/\theta)\big).$$

The color multiplicity on one directed pair is $(2 + o(1))M/(r^2 D)$, and the working precision is $(6 + o(1))MD/u$. Applying the generating-function bound with tilt θ/z_e, summing with weight z_e, and optimizing the leading terms gives $\pi\sqrt{2}M\|z\|_2/(r\sqrt{u})$ in natural logarithms. Division by $\ln 2$ gives (2.2).

2.2 Paid recursive calls

The first stage caches each Z_g on all incident pair indices. Its incident set has $2(r-1)$ directed pairs, so the total number of child calls has exponent at most $(\alpha_r + o(1))\mathcal{K}$, where

$$\alpha_r = \frac{\sqrt{2(r-1)}}{r}. \tag{2.3}$$

A local contraction plan then combines these cached functions. If its certified nonrecursive time coefficient is β_r, with backend slack $\delta > 0$ charged explicitly, the recurrence is

$$T_d \le \mathrm{poly}(s)\left[2^{(\alpha_{r_d} + o(1))\mathcal{K}_d} T_{d+1} + 2^{((1+\delta)\beta_{r_d} + o(1))\mathcal{K}_d}\right]. \tag{2.4}$$

The local space coefficient is at most β_{r_d}; the input/output term of the matrix cost pays for every table. Recursive tables on a stack add in space, whereas ancestor child-call exponents add in time. Ignoring that distinction would give an incorrect headline bound.

2.3 What an improved wire coefficient means for SAT

All numerical coefficients in this report multiply $E(s)$ per *normalized Boolean wire*. The known reduction yields, for fixed k, the inherited density-sensitive order

$$\mathrm{poly}_k(n,m)\,2^{O_k\left(\frac{n}{\sqrt{\log n}}\min\{\log(2+m/n),\log\log n\}\right)}. \tag{2.5}$$

The direct clause-count branch and all stronger-assumption low-degree branches remain available. Smaller explicit wire coefficients give smaller certified contraction upper bounds. They do not determine numerical coefficients per SAT variable without the reduction's wire constant, and they do not change the logarithmic order in (2.5).

3 An exact minimax theorem for a finite menu

The central new tool is a characterization of the exact worst-case cost of a fixed finite menu of contraction plans. Earlier support certificates average the plans with equal weights. That is a valid upper bound, but need not be the optimum certificate for the same menu.

Theorem 3.1 (Exact support-hull identity)**.** *Let $\mathcal{V}_1,\dots,\mathcal{V}_q$ be nonempty finite subsets of $\mathbb{R}^E_{\geq 0}$, where E is finite. Define*

$$F(g)=\min_{1\leq j\leq q}\max_{v\in\mathcal{V}_j} v\cdot g, \qquad \Lambda=\max_{v_j\in\mathcal{V}_j}\operatorname{dist}\bigl(0,\operatorname{conv}\{v_1,\dots,v_q\}\bigr).$$

Then

$$\max_{g\geq 0,\,\|g\|_2\leq 1} F(g)=\Lambda. \tag{3.1}$$

Equivalently,

$$\Lambda^2=\max_{v_j\in\mathcal{V}_j}\ \min_{p\geq 0,\,\sum_j p_j=1} p^TGp, \qquad G_{ij}=v_i\cdot v_j.$$

For rational supports, Λ^2 is rational and admits a finite exact certificate consisting of rational convex weights and a projection witness.

Proof. Fix $g\geq 0$ with $\|g\|_2\leq 1$. Select independently for each plan a support v_j attaining its maximum. If w is the point of the convex hull of these supports closest to zero, then

$$F(g)=\min_j v_j\cdot g\leq w\cdot g\leq\|w\|_2\leq\Lambda.$$

Conversely, choose a support tuple attaining Λ and its closest point w. The projection condition gives $(v_j-w)\cdot w\geq 0$ for every j. If $w=0$, the reverse inequality is automatic. Otherwise take $g=w/\|w\|_2$. Nonnegativity of the supports makes g feasible, and

$$\max_{v\in\mathcal{V}_j} v\cdot g\geq v_j\cdot g\geq\|w\|_2=\Lambda \quad\text{for every } j.$$

This proves equality. The Gram expression is the definition of the distance to a convex hull.

The closest point lies in the relative interior of a simplex spanned by an affinely independent subset of the rational vertices. Its affine projection solves a rational linear system, so its coordinates and squared norm are rational. One can enumerate faces to find it. The conditions below give an exact verification that does not depend on numerical optimization. □

3.1 A small exact certificate

For a proposed nonempty active set $A \subseteq \{1, \dots, q\}$, solve

$$\begin{pmatrix} G_{AA} & -\mathbf{1} \\ \mathbf{1}^T & 0 \end{pmatrix} \begin{pmatrix} p_A \\ t \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \tag{3.2}$$

Use $p_j = 0$ off A. A solution certifies a closest point if

$$p \geq 0, \quad \mathbf{1}^T p = 1, \quad Gp \geq t\mathbf{1}, \quad (Gp)_j = t \ \ (p_j > 0).$$

Indeed $p^T Gp = t$, and for any other simplex vector z,

$$z^T Gz = p^T Gp + 2(z-p)^T Gp + (z-p)^T G(z-p) \geq t.$$

The last term is nonnegative because G is a Gram matrix. Singular active sets can be replaced by an affinely independent subset representing the same closest point. All checks are rational equalities or inequalities.

For an upper certificate it is even simpler: *any* rational simplex vector p with $p^T Gp \leq B$ proves that tuple's distance squared is at most B. A projection witness attaining B proves sharpness. Our checker first uses $p_j = 1/q$. Only tuples for which the equal-average bound exceeds the candidate threshold require the more expensive exact face calculation. This makes exhaustive certification practical.

3.2 Application to contraction plans

Combine opposite directed entropies as $g_{\{a,b\}} = h_{ab} + h_{ba}$. Suppose every support of plan j is an integer vector v and its operation exponent is $d^{-1} \max_{v \in \mathcal{V}_j} v \cdot g$. If $R = \Lambda^2$ is the certificate from Theorem 3.1, its local normalized coefficient is

$$\beta_r = \frac{\sqrt{2R}}{dr}. \tag{3.3}$$

To see this without assuming uniformity over infinitely many test vectors, fix one support tuple and its rational closest combination w. Give both orientations of each pair weight w_e/d. The squared directed norm is $2\|w\|_2^2/d^2$. Apply the inherited weighted entropy estimate to this vector. There are finitely many tuples, so its lower-order error is uniform over every vector used in this argument. Select the plan minimizing its actual operation exponent.

The former equal-average certificate is recovered by always taking $p_j = 1/q$. Thus the new theorem is never weaker for a fixed menu. It is strictly stronger when the worst applicable tuple has a closer nonuniform convex combination. It also supplies a matching worst-case entropy vector, so one can distinguish a poor proof of a menu from a menu that must actually be enlarged or changed.

Scope of sharpness. Equation (3.1) is exact for the nonnegative Euclidean entropy relaxation. It does not assert that every maximizing vector g comes from the exact weight lists of an actual Cauchy contraction. It is therefore a sharp finite certificate for the stated model, not a lower bound for SAT, for all contraction algorithms, or for the true runtime of a plan.

Exact online selection. Let $D_{ab} = |\mathcal{I}_{ab}|\,|\mathcal{I}_{ba}|$. For each plan compute

$$P_j = \max_{v \in \mathcal{V}_j} \prod_{\{a,b\}} D_{ab}^{v_{ab}}.$$

The plan minimizing P_j also minimizes the real operation exponent. All dimensions are positive integers; empty-list allocations contribute zero and are skipped. The fixed supports have constant integer entries, and the dimension counts have polynomial bit length, so this selection has polynomial bit cost and requires no floating-point comparisons.

4 Two exact matrix backends

We give two versions of the result. The first uses the classical field bounds already available to the incoming rectangular packet. The second also imports the stronger published 2026 matrix-multiplication theorems. Keeping both versions makes the additional dependency explicit.

4.1 A common rectangular formula

Suppose fixed bilinear matrix algorithms are available with dual aspect ratio $a \in (0,1)$ and square exponent $w > 2$, allowing arbitrarily small positive exponent slack. For sorted base-two logarithmic dimensions $x \geq y \geq z \geq 0$, define

$$B_{a,w}(x,y,z) = \max\left\{x + y, \quad x + \frac{1 - a(w-1)}{1-a}y + \frac{w-2}{1-a}z\right\}. \tag{4.1}$$

For every fixed $\delta > 0$, there is a fixed exact algorithm with time a polynomial factor times $2^{(1+\delta)B_{a,w}}$. Its space is a polynomial factor times the largest input or output array.

Indeed, block the largest dimension at scale 2^y. For $z \leq ay$ use the dual-exponent algorithm. Otherwise tensor the dual and square schemes at logarithmic scales

$$t_0 = (y - z)/(1 - a), \qquad t_1 = (z - ay)/(1 - a).$$

They satisfy $t_0 + t_1 = y$ and $at_0 + t_1 = z$. Their rank exponent, plus the blocking cost, is exactly the second branch of (4.1). Fixed-power padding changes sizes by constant factors. Choosing each component's exponent slack sufficiently smaller than 2δ gives the stated multiplicative allowance, since $B_{a,w} \geq x + y \geq 2y$. The first branch is the essential input/output cost.

For the classical version use $a = 3/10$ and $w = 19/8$, justified by the strictly stronger results of Vassilevska Williams, Xu, Xu and Zhou [13]. Then

$$B_{\mathrm{cl}} = \max\{x + y,\ x + (47/56)y + (30/56)z\}. \tag{4.2}$$

For the additional 2026 version use $a = 9/20$ and $w = 9/4$. The published results give $\omega(\mathbb{C}) \leq 9/4$ and a characteristic-zero dual exponent greater than 0.465, hence greater than $9/20$ [14, 15]. Therefore

$$B_{\mathrm{new}} = \max\{x + y,\ x + (35/44)y + (20/44)z\}. \tag{4.3}$$

We fix $\delta = 10^{-5}$ in the final schedules. This charges the positive slack of the algorithms explicitly. The endpoint statement $\omega \leq 9/4$ by itself is not being interpreted as one fixed algorithm of exact cost $N^{9/4+o(1)}$.

4.2 Transfer from complex arithmetic to exact bit arithmetic

The stronger result is stated over $\mathbb{C}$, whereas the SAT contractor uses rings with zero divisors. The transfer requires an argument.

Fix the positive tolerance once. The matrix theorem supplies a finite exact rank scheme. Its polynomial equations have rational coefficients; if they have a complex solution, they have a solution over $\overline{\mathbb{Q}}$. All coefficients of this fixed scheme lie in a single number field $K/\mathbb{Q}$ of fixed degree. The square paper explicitly records this observation, and the rectangular paper states its characteristic-zero transfer [14, 15]. Choose the finitely many schemes used by the two tensor factors in a common number field.

Represent elements of K in a fixed rational basis. A field operation becomes a fixed number of rational-coordinate operations. Recursive bilinear schemes form only fixed linear combinations of input blocks until their leaf products, then fixed linear combinations of outputs. Consequently their intermediate coordinate bit lengths are polynomial in the input bit length and the logarithm of the matrix dimensions. Fixed denominators and coefficient heights accumulate over only logarithmically many recursive levels. This gives a uniform exact rational implementation with the same arithmetic exponent and polynomial bit overhead.

Now write an entry of the SAT coefficient ring in its polynomially many integer module coordinates. Lift the coordinates from $\mathbb{Z}/2^p\mathbb{Z}$ to $[0, 2^p)$. For each pair of module coordinates, compute the corresponding integer matrix product using the rational implementation. The resulting entries have bit length $O(p + \log d_2)$ for inner dimension d_2. Combine them using the ring's integer multiplication table and reduce modulo 2^p. All ring dimensions, precisions, and logarithmic matrix dimensions are polynomial in s. Coordinate pairs can be processed sequentially. This proves the required bit-cost and storage bounds.

Alternatively, compute those integer products modulo enough good odd primes and reconstruct them by CRT, specializing the fixed number field through a fixed-degree quotient algebra. Only finitely many denominator primes are excluded. No assertion about a dual exponent directly in characteristic two is needed, and no nonunit is inverted in the SAT ring.

The fixed number-field schemes are existence ingredients of the asymptotic algorithm. The delivered finite graph evaluator uses ordinary exact multiplication; it does not implement these large fast schemes or claim a practical crossover improvement.

4.3 Operation supports

For a merge of disjoint vertex sets A, B, let $O = [r] \setminus (A \cup B)$. The three matrix axes are the independent pair-index classes A–B, A–O, and B–O. The final triangle is treated in the same way. Since the coefficients in either nontrivial branch above are nonincreasing, the rearrangement inequality expresses the sorted cost as the maximum over permutations of its coefficients. Thus the classical integer supports are the permutations of

$$(56, 56, 0), \quad (56, 47, 30),$$

with divisor 56, and the additional backend uses

$$(44, 44, 0), \quad (44, 35, 20),$$

with divisor 44. There are nine distinct coefficient triples in each case. An edge internal to one part gets coefficient zero; otherwise its coefficient is that of its pair class. The support sets include every binary merge and the final triangle of each plan.

5 Explicit contraction menus and their certificates

A plan is three ordered blocks. Inside each block, split the ordered list at $\lfloor k/2 \rfloor$, build the two tables recursively and merge them. Finally contract the three blocks. There are $r-2$ operations and therefore $9(r-2)$ supports per plan.

The nine-group menu is the four parallel classes of the affine plane $\mathbb{F}_3^2$, with labels $0, \dots, 8$ in row-major order:

$$\begin{array}{lll} (0,1,2) & (3,4,5) & (6,7,8) \\ (0,3,6) & (1,4,7) & (2,5,8) \\ (0,4,8) & (1,5,6) & (2,3,7) \\ (0,5,7) & (1,3,8) & (2,4,6). \end{array}$$

The twelve-group menu is

$$\begin{array}{lll} (0,1,2,3) & (4,5,6,7) & (8,9,10,11) \\ (9,1,7,0) & (6,11,2,4) & (5,3,8,10) \\ (4,10,0,7) & (11,6,8,1) & (2,3,9,5) \\ (7,5,3,11) & (9,4,8,0) & (6,1,2,10). \end{array}$$

The optional ten-group menu is

$$\begin{array}{lll} (0,1,2,3) & (4,5,6) & (7,8,9) \\ (2,6,3,7) & (0,5,8) & (4,9,1) \\ (4,8,3,2) & (1,5,7) & (9,0,6) \\ (3,1,8,6) & (0,4,7) & (9,5,2). \end{array}$$

These explicit menus, rather than the heuristic search used to suggest two of them, constitute the mathematical construction.

Backend and groups	Supports/plan	Tuples covered	Certified β^2
Classical, $r = 9$	63	15,752,961	54113/169344
Classical, $r = 12$	90	65,610,000	4110726305041430303/12559431766773663744
Newer, $r = 10$	72	26,873,856	119061/387200
Newer, $r = 12$	90	65,610,000	55241/185856

The first three bounds are sharp for their finite menus under the Euclidean relaxation. The final entry is a sufficient uniform-average upper bound. For the classical twelve-group menu the sharp hull squared norm is

$$R = \frac{12332178915124290909}{166871701833196}.$$

Only 18 tuples require a nonuniform hull calculation after the uniform average pruning. The maximizing Gram matrix is

$$\begin{pmatrix} 99920 & 65557 & 64979 & 64511 \\ 65557 & 99920 & 65395 & 65557 \\ 64979 & 65395 & 99920 & 65395 \\ 64511 & 65557 & 65395 & 99920 \end{pmatrix},$$

and its optimal weights have common denominator 166871701833196 and numerators

$$42504471540671, \quad 40721557063169, \quad 41630292075405, \quad 42015381153951.$$

The independent unpruned replay agrees with the production certificate, which additionally uses exact integer branch bounds to cover the complete Cartesian products efficiently. The production checks cover 173,846,817 support tuples in all four cases, rather than sampling entropy vectors.

Plan selection itself uses no real-number decisions. For each support v compute the integer product $\prod_{g<h}(|\mathcal{I}_{gh}|\,|\mathcal{I}_{hg}|)^{v_{gh}}$; minimize the largest such product over the four plans. The fixed dimensions and polynomial-bit list cardinalities make this preprocessing polynomial in the original size parameter.

6 A fully paid multistage schedule

Let $\lambda = \lceil \log_2 s \rceil$, use $u_d = \lfloor \eta_d \lambda \rfloor$, and put

$$a_d = \sum_{j \le d} u_j + (d+1)\lceil 6 \log_2 \lambda \rceil, \qquad H = \lceil s/\lambda \rceil.$$

Stop when the child wire count is at most H; use direct contraction for bounded exceptional sizes or failed finite admissibility tests. The classical prefix is

$$(r_0, \dots, r_7) = (9, 12, 8, 6, 5, 4, 4, 4),$$

$$(\eta_0, \dots, \eta_7) = (0.01577, 0.01212, 0.00544, 0.00535, 0.00438, 0.00342, 0.00215, 0.00136).$$

Every displayed decimal is an exact terminating rational. For $d \ge 8$ take

$$r_d = 3, \qquad \eta_d = \frac{30107}{10^7}\left(\frac{77}{100}\right)^{d-8}.$$

The total retained fraction is $1577/25000 = 0.06308$. For the optional newer backend replace only

$$r_0 = 10, \qquad \eta_0 = 0.01505, \qquad \eta_1 = 0.01284.$$

Both prefixes have cumulative range 0.02789 at depth one and 0.04999 at depth seven.

Let $A_d = \sum_{j \le d} \eta_j$. At the root the strict leading packing gate is $9/10 + 2A_0 \log_2 r_0 < 1$. Later gates are

$$9/10 + A_d \log_2 r_d < 1, \qquad 2A_d \log_2 r_d + A_{d-1} < 1, \qquad (2\log_2 r_d + 1)A_d < 1.$$

The accompanying interval checker proves all of them, including their tail limits, by exact rational enclosures of the logarithms. The strict positive margins absorb all polylogarithmic padding and precision factors. All fresh tensor degrees are at most $s^{0.06308+o(1)} < s^{9/10}$.

For a noncircular depth argument, first verify these inequalities through $\log_{600/431} \lambda + O(1)$ hypothetical active levels. The minimum taper is at least $\lambda^{1+\ln(77/100)/\ln(600/431)+o(1)}$, which diverges. Nearly lossless padding then gives eventual tail child ratio at most $431/600$, since its sharp ratio is $2/3$, closing the depth bootstrap. Padding tolerance $O(\lambda^{-3})$ makes the accumulated relative padding error $1 + o(1)$. The six-logarithm buffer absorbs inherited precision and gives the uniform local leading precision $(6 + o(1))MD/u$.

Every group cache calls its true child contractor. Thus the paid recurrence is

$$T_d \le \mathrm{poly}(s)\left[2^{(\alpha_{r_d}+o(1))K_d} T_{d+1} + 2^{((1+\varepsilon)\beta_{r_d}+o(1))K_d}\right], \qquad \alpha_r = \frac{\sqrt{2(r-1)}}{r}.$$

Use the certified menus at the first two depths and fixed balanced three-block trees thereafter. The latter have

$$\beta_8^2 = 15/32, \quad \beta_6^2 = 4/9, \quad \beta_5^2 = 12/25, \quad \beta_4^2 = 1/2, \quad \beta_3^2 = 4/9$$

under either backend. Their input/output support bounds dominate the backend-dependent arithmetic supports.

Put $m_0 = 1$, $m_d = \prod_{j<d} 2/r_j$, and $K_d^* = (\pi\sqrt{2}/\ln 2)m_d/\sqrt{\eta_d}$. The prefix branch at depth d has coefficient

$$\sum_{j<d} \alpha_{r_j} K_j^* + (1+\varepsilon)\beta_{r_d} K_d^*.$$

For the tail use $\gamma = (2/3)/\sqrt{77/100}$ and append

$$\sum_{j<8} \alpha_{r_j} K_j^* + K_8^* \max\{(1+\varepsilon)\beta_3, \alpha_3/(1-\gamma)\}.$$

The space coefficient is $\max_d \beta_{r_d} K_d^*$: live ancestor and child tables add in storage rather than multiply. Exact rational interval arithmetic gives

	C_T	C_S
Classical backend	30.0872056033774679…	28.8528994149423854…
Optional newer backend	28.9729416032324588…	28.9726518767136916….

Therefore eventual coefficients $(30.09, 28.86)$ and $(28.98, 28.98)$ are valid. The classical time maximum occurs at depth one; the optional time maximum occurs at the root. Neither time statement uses polynomial space.

Direct leaves cost $2^{O(s/\lambda)}$; permitted original entries cost $2^{O(s^{9/10})}$ times a fixed polynomial; fresh entries cost $\exp(s^{0.06308+o(1)})$. These are lower order after all ancestor calls have been charged. Polynomial work over $O(\log\lambda)$ levels adds $O(\lambda\log\lambda)$ to the exponent. Ring dimensions, precisions and record lengths stay polynomial in s. Sorting and scanning the tables in index order retains the reviewed bit-cost model.

6.1 Fair comparisons and the remaining goal

All rows below use the same normalized-wire interface. "Old" means the reviewed three-plan method of [9], not an arbitrary optimization over all previous methods. Backend slack is paid in the newly composed constructions.

Construction	C_T	C_S
Best displayed reviewed cached proposal	30.891019	30.753913
Old menus, classical rectangular backend	30.629542	29.985632
Old menus, classical backend, retuned tail	30.615144	29.966611
New menus, classical backend	**30.087206**	**28.852899**
Old menus, newer backend, retuned tail	30.273960	28.625470
New menus, newer backend	**28.972942**	**28.972652**

On the old published schedule, merely inserting the newer backend gives 30.512927432781… time; the explicit tail retuning gives the 30.273960451080… row. Thus the new time improvement survives equal backend upgrades. That retuned old-menu row has a smaller space coefficient than the new fastest construction; retain it when memory is the priority. These are explicit compatible comparisons, not a global optimization of every possible composition of previous algorithms.

The new estimates remain $\mathrm{poly}(s+\mathcal{L})2^{(C+o(1))s/\sqrt{\log_2 s}}$. They do not improve the worst-case SAT order $n\log\log n/\sqrt{\log n}$, prove the conditional $n/\log n$ interfaces, or supply a complete SAT solver. The polynomial-space frontier $46.184845586731\ldots$ remains a separate resource option. The exact minimax identity and its certificate compiler are new mathematical tools for obtaining the best constant of a fixed finite strategy menu; they do not by themselves remove the square-root logarithm.

7 Attempts to change the logarithmic order

The numerical improvement is useful, but the requested larger goal is a smaller order than $n\log\log n/\sqrt{\log n}$. We investigated the following routes and retain their precise status. None of the arguments below proves a lower bound for SAT.

7.1 What is already ruled out by the reviewed certificates

Changing the Hermite taper alone is not a new route. The earlier precision packet solves the real taper minimax problem, and the newest rectangular packet strengthens this to arbitrary asymmetric node multiplicities under independent jet-error recovery. Its leading precision coefficient remains six. Similarly, full-block and actual-domain Cauchy/Smith separation is already in the corpus. Merely rediscovering the slope-four mode list is not an improvement.

The conserved degree-flow identity in the newest adaptive-frontier packet is useful: incoming mode powers remove the degree allowance that outgoing mode powers add elsewhere. But a whole-group coefficient extraction can still be expensive. The same packet explicitly distinguishes formal degree conservation from an admissible recursive entry algorithm. A short binary rank for a mode tuple similarly does not make a high-degree group tensor cheap to evaluate.

7.2 Why straightforward composition of the two engines cancels

Here is a quantitative obstruction for the displayed certificates, not for all possible algorithms. Suppose an OpenAI packing schedule obeys

$$u\log_2 r \le B\log_2 s,$$

and its certified outer mode-bit size is $AM/\sqrt{u}$. Suppose the Anthropic clique certificate applied to r groups has coefficient at least $a/\sqrt{\log_2 r}$, as does the current square-root family up to its fixed constants. Their composed *certificate expression* is at least

$$\frac{aAM}{\sqrt{u\log_2 r}} \ge \frac{aA}{\sqrt{B}}\frac{M}{\sqrt{\log_2 s}}.$$

Thus allowing r to grow, while retaining these two separate estimates, does not yield a smaller order. It loses $\sqrt{\log r}$ in packing and gains the same factor in clique evaluation. This argument is about those upper-bound expressions; it does not lower-bound their algorithms' true running time or exclude a joint analysis that avoids one of the charges.

There is a related exact calculation for rectangular grids in the Anthropic construction. Its two intermediate partition-space envelopes are proportional to $\sqrt{2h+t}$ and $\sqrt{2h-t}$. If the two recursive directions have rates e_1, e_2, then

$$\max_{-2h\le t\le 2h}\min\{e_1\sqrt{2h+t}, e_2\sqrt{2h-t}\} = \frac{2\sqrt{h}}{\sqrt{e_1^{-2}+e_2^{-2}}}.$$

The maximum occurs where the two terms agree. After normalization, the combined rate is $(e_1^{-2} + e_2^{-2})^{-1/2}$. If $e(K) = C/\sqrt{\log K}$ and the grid combines sizes as $K = K_1 K_2$, this is exactly the same rate $C/\sqrt{\log K}$. Unequal grid sides alone therefore do not change the power within this envelope.

7.3 A more concrete new route: compile the actual piece ranks

The Anthropic proof bounds the ranks of a completed crossing piece by the dimensions of intermediate Fock spaces. The completed matrix could in principle have smaller rank after cancellations. Rather than changing the Heisenberg algebra or its grading, one can ask directly for a uniform rank theorem for those matrices. The accompanying `research/MODULAR_RANK_CRITERION.md` supplies the full statement and proof obligations; we give the central lemma here.

Let $N = 2^\ell$ and suppose polynomially many integral arrays $A_i(x, y, u, v)$ satisfy

$$\sum_i A_i(x, y, u, v) = d_\ell[x = v][y = u],$$

with polynomial coefficient bit lengths and $2^{O(\ell)}$ entry cost. For a prime $p \nmid d_\ell$, flatten a piece in the row direction $(x, y) \mid (u, v)$ and the column direction $(x, u) \mid (y, v)$ over $\mathbb{F}_p$. Denote their ranks by $r_i(p), c_i(p)$.

Lemma 7.1 (Modular rank compilation)**.** *Both exact factorizations of a piece can be constructed and verified in* $2^{O(\ell)} \operatorname{poly}(\log p)$ *bit operations and space, using respectively*

$$\lceil \log_2 \max(1, r_i(p)) \rceil, \qquad \lceil \log_2 \max(1, c_i(p)) \rceil$$

shared Boolean bits. Every factor coefficient has at most $\lceil \log_2 p \rceil$ *bits.*

Proof. Each flattening is an N^2 by N^2 matrix. Gaussian elimination over $\mathbb{F}_p$ takes $O(N^6)$ field operations. Its pivot columns and the corresponding reduced-row coefficients give an exact rank factorization UV. Pad the inner dimension to the next power of two and encode it by Boolean bits. Zero rank uses one zero term and no index bits. Multiplying U, V verifies the factorization within the same time bound. Field arithmetic has polynomial bit cost in $\log p$. Reduction preserves the crossing identity and $p \nmid d_\ell$ permits its normalization. □

This modular formulation avoids a real height gap. A generic rational Gaussian elimination proof only gives minor determinants with $2^{O(\ell)}$ bits; it does not automatically meet the source's polynomial coefficient-bit requirement. Modular factor coefficients remain small. The lemma is not a polynomial-space claim in ℓ.

The substantive *open* target is, for some fixed $1/2 \le \rho < 1/\sqrt{2}$, the uniform inequality

$$\lceil \log_2 \max(1, r_i(p)) \rceil + \lceil \log_2 \max(1, c_i(p)) \rceil \le 4\rho\ell + O(\log(2\ell)). \tag{7.1}$$

It must hold for every needed piece and for a polynomially constructible family of good primes with the signed-reconstruction size specified below. For the original normalizer $d_\ell = h!$, taking primes above every reached h avoids division problems. The number of pieces must remain polynomial in ℓ, and the constants must be uniform; a few favorable pieces or a single exceptional prime are not sufficient.

A rational-rank upper bound is especially useful: every larger minor is an identically zero integer, hence vanishes modulo every prime. Thus one may prove the rank bound over $\mathbb{Q}$ and construct the small-coefficient factors over finite fields. A concrete intermediate target is

$$\max(1, \operatorname{rank}_{\mathbb{Q}} R_i) \max(1, \operatorname{rank}_{\mathbb{Q}} C_i) \le 2^{(8/3)\ell} \ell^{O(1)}$$

uniformly for all pieces. This would give $\rho = 2/3$ and denominator power $\gamma = \log_2(3/2) = 0.5849625\ldots > 1/2$.

If (7.1) and the stated uniform interfaces are established, the source recurrence gives $\gamma = \log_2(1/\rho) > 1/2$ and the corresponding fixed-width SAT exponent $O_k(n \log\log n/(\log n)^\gamma)$. There is an essential arithmetic detail: Appendix D uses truncation congruences modulo powers of two. It cannot simply be rerun in odd characteristic and followed by a CRT of the final count.

Instead supply it with an exact integer clique oracle. For a (K, ℓ)-clique with local weights satisfying $|W_v| < 2^B$, its value Z obeys

$$|Z| \le 2^E, \qquad E = \binom{K}{2}\ell + KB.$$

Compute the requested clique value modulo good primes whose product is strictly greater than 2^{E+1}, then reconstruct its signed integer value. The original two-adic truncation and rational comparisons now run unchanged. It is enough to reconstruct each clique value requested by the outer reduction; internal recursive modular values need no such recovery. The source's parameter bounds make K, ℓ, B, and hence E, polynomial in the original input size. Polynomially many polynomial-size primes suffice and do not change the exponential order. The direct clause-count branch, which counts directly modulo each prime, needs only the final nonnegative count bound 2^n.

The rank theorem is still missing. This is a *conditional route*, not a newly proved SAT bound. The accompanying note states the piece-count, normalization, construction-time and coefficient-height obligations in full so that a rank conjecture cannot silently replace an algorithm.

7.4 Finite probes and their limitations

Exact modular probes of the actual Boson pieces did not reveal a uniform collapse. For example, central-piece row/column ranks over $\mathbb{F}_{1009}$ were $(25, 25)$ for the full weight-4 sector, $(49, 49)$ for weight 5, and $(225, 225)$ for weight 7. Their dimensions are respectively 5^2, 7^2, and 15^2, so these are full-rank finite witnesses. Full modular rank also proves full rational rank for those particular integer matrices. At these sizes the source's partition-dimension bound has not entered its asymptotically useful range, so the observations neither prove nor refute (7.1).

The standalone rank compiler was separately checked on exact factorizations, crossing identities, exceptional-prime controls, and CRT reconstruction. An explicit negative control uses matrix-unit crossing pieces with ideal individual ranks but exponentially many pieces; it fails the source interface and cannot be promoted to a faster SAT theorem.

Borchardt's Cauchy-permanent identity is another possible algebraic tool, but applying it only to one packet would speed a subdominant local routine. A useful smaller-order result would still require a closed group-level representation with paid entry evaluation. We did not establish that representation or use this idea to claim an improved exponent.

8 What the supplied Anthropic commit changes

The link supplied during this review is pull request 31 at commit `1a6eee293b9e` of `anthropics/formal-math` [16]. It publishes the paper and Lean development of *A $2^{o(n)}$ Time Algorithm for 3-SAT*. This supersedes the older ProveIt overview's statement that the development had not been published. The closing commit `7de9d09fecd1` already corrects that overview and links this development.

8.1 The exact public endpoint

The file `eth/MainResults.lean` imports Mathlib alone and defines the public statements. Its main theorem asserts the existence of one deterministic finite-alphabet Mathlib multi-stack Turing machine and one constant C, both chosen before the clause width k, such that the machine counts satisfying assignments in at most

$$2^{C \min\{km/\sqrt{\log_2 m},\ k^2 n \log_2 \log_2 n/\sqrt{\log_2 n}\}} \tag{8.1}$$

steps, with the stated integer rounding. Its premises are $k \geq 1$, $n \geq 4$, $m \geq 2$, at most k literals per clause, no repeated clause, and occurrence of every variable. Its output is the exact count in $n+1$ binary digits. The theorem about ETH is stated separately and follows through explicit front ends.

The proof in `eth/Solution.lean` invokes `CountedRoute.mainTheoremHolds`. The separate ETH endpoint is constructed through `NamesInOrder` and `Wrap`. The two `sorry` declarations in `eth/Challenge.lean` are intentionally unproved comparison statements. The solution does not import that challenge file.

The pinned checkout contains 1,052 Lean source files. A static scan found the two intentional challenge placeholders and no other literal `sorry/admit` declaration. That scan is not a kernel proof check. The README reports the build, axiom listing with `propext`, `Classical.choice`, and `Quot.sound`, and the comparator/nanoda workflow. We inspected the source and entry points; we did not independently rebuild the development or reproduce those reported kernel checks in this session.

The supplied 49-page PDF is not byte-identical to the earlier 49-page `eth.pdf` in ProveIt. Both hashes are retained in the source ledger. The algorithmic order in (8.1) agrees with the previous manuscript's stated order. Publication of the development is an important change in what can be inspected, rather than a new smaller asymptotic order to beat.

8.2 Why its constants cannot simply replace wire constants

The solution comments explicitly distinguish intermediate counted operations from final Turing-machine steps. The simulation can multiply the intermediate exponent by an unspecified constant, absorbed in the big-O theorem. Consequently an internal clique coefficient or rank ratio is not a numerical competitor to the Cauchy contractor's $C\,s/\sqrt{\log_2 s}$ bit-time coefficient.

The two algorithms nevertheless suggest a useful research comparison. The OpenAI construction organizes a weighted Cauchy mode sum, whereas the Anthropic construction rewrites crossings with operators on a graded polynomial space and then selects a contraction direction. Both currently lead to a square-root logarithm. Section 7 and its accompanying research notes test whether their structures can be combined without hiding entry costs or paying the gain back in a different parameter.

9 Validation, reproducibility, and remaining proof obligations

9.1 What was checked

The delivered production verifier constructs every support from the stated ordered plans and matrix weights. Rigorous integer branch bounds cover 173,846,817 Cartesian support tuples, directly evaluate 7,560 tuple norms, and solve 18 nonuniform hull problems with exact rational arithmetic. Every pruned block is bounded above, counted, and included in an exact sum-of-squared-norms

checksum. The code validates the projection inequalities for the sharp witnesses. It does not sample a continuous set of entropy vectors or use floating-point optimization as a proof.

The independent implementation reconstructs the supports and evaluates all tuple norms without the production branch pruning. It obtains the same tuple counts, maxima, checksums and certified rational bounds. Its support ordering may differ, so witness indices need not be identical. The twelve-group certificate for the newer backend is an upper bound; only the other three certificates are asserted to be sharp for their menus.

The schedule verifier uses convergent rational series for logarithms and π, and integer square-root enclosures. It checks every prefix gate, the infinite-tail gates, the complete paid time recurrence, storage, and strict separation from the rounded eventual constants. No binary floating-point result is used to accept a schedule. The independent schedule check also verifies the classical packing gates by integer-power comparisons and recomputes its constants at high precision.

A small exact evaluator checks all four plans on sixteen finite mode-graph fixtures over $(\mathbb{Z}/64\mathbb{Z})[V]/(V^2)$, including a zero divisor and a nonzero nilpotent. Its 64 plan evaluations agree with literal enumeration over 1,024 assignments. It records 495 actual child-entry calls while constructing the caches, and checks two invalid-input controls. This tests the contraction and selection identities over a nonfield; it is not an implementation of the asymptotically fast matrix backend.

The modular-rank compiler passes 945 finite-field factorization cases, nine crossing-identity cases, four invalid-input rejections, an exceptional-prime rank check, 2,049 nonnegative CRT cases, 4,097 signed CRT cases, and 50 signed-network cases followed by two-adic reduction. One negative control demonstrates why the least nonnegative CRT representative cannot simply be reduced modulo 2^ν. Another demonstrates that exponentially many small-rank pieces violate the uniform polynomial-piece interface.

All delivered production drivers were run both normally and with `python3 -O`; their mathematical results agree. These checks use explicit exceptions rather than removable assertions. The full replay, artifact hashes and dependency boundaries are recorded in `results/` and `PROOF_STATUS.json`.

9.2 How to reproduce the finite claims

From the archive's top-level directory, run

```
python3 run_checks.py
python3 run_checks.py --optimized
python3 audit/independent/replay_menu_certificate.py --help
python3 build_report.py
python3 verify_bundle.py
```

The core certificate, schedule, comparison, algebra and modular checks use Python's standard library. The optional independent exhaustive replay uses NumPy; its independent high-precision schedule check uses mpmath. The PDF build requires a standard TeX installation with pdfLaTeX and the packages listed in `report.tex`. The exact command lines and expected outputs are in `README.txt`.

9.3 What has not been established

The new finite minimax theorem, menu certificates, matrix-coordinate transfer argument and schedule analysis are supplied here. The exact deformation, Hermite recovery, Cauchy mode identity, entry interfaces and SAT reductions remain imported from the cited sources. This is therefore a relative improvement of those constructions, with those premises made explicit, rather than an independent proof of their entire SAT theorem.

We did not build a complete SAT solver, benchmark a SAT instance, implement the large fixed fast-matrix schemes, or independently rebuild the supplied Lean formalization. No claim of practical crossover, global optimality over all contraction plans, polynomial-space improvement, or a smaller worst-case logarithmic order is made. The actual-rank criterion in Section 7 is an open theorem obligation, not a consequence of the small finite probes.

The archive is suitable for review and integration as a new research packet. Its exact finite certificates and explicit assumptions make the proposed improvement concrete; any future claim of a better logarithmic power should discharge the rank criterion or supply a different paid uniform interface.

References

[1] Vladimir Reshetnikov, *ProveIt: SubexponentialSAT*. Initial snapshot `73dfb5e5ab6e`; closing snapshot `7de9d09fecd1`. Full hashes are in the provenance ledger. Closing project layout. Placed reports; nine incoming archives.

[2] OpenAI, *The Exponential Time Hypothesis is False*, October 25, 2026 revision. `openai/math@7ef5c0cc5c`. Paper and sources.

[3] OpenAI, quantitative tensor-contraction appendix, `build/quantitative-contraction.tex`, same pinned preprint. Exact deformation, recovery, recursion, and bit-cost interface.

[4] *Sharper precision for subexponential 3SAT*, ProveIt packet `3sat_precision_refinement`, October 26, 2026. Report source.

[5] *Asymmetric packet balancing*, ProveIt packet `eth_asymmetric_packet_balancing`, October 26, 2026. Report source.

[6] ProveIt packet `eth_joint_cauchy`, October 26, 2026. Full-block Cauchy modes and integral factorization.

[7] ProveIt packet `sat_adaptive_contraction`, October 26, 2026. Report source.

[8] ProveIt packet `subsat_contraction_tree`, October 26, 2026. Paid cached-contraction interface and ring bridge.

[9] *Adaptive Contraction Trees with Exact Support Certificates*, incoming `sat_adaptive_tree_certificate_2026-10-26.zip`. Pinned archive.

[10] *Adaptive Contraction Beyond the October 26 SAT Frontier*, incoming `sat_adaptive_bounds_2026-10-26.zip`. Pinned archive.

[11] *Rectangular Contraction, Streamed Conditioning, and Precision Barriers*, incoming `subsat_rectangular_2026-10-26.zip`. Pinned archive.

[12] ProveIt packets `sat_density_improvement` and `eth_density_refinement`, October 26, 2026. Density-sensitive reduction and report.

[13] Virginia Vassilevska Williams, Yinzhan Xu, Zixuan Xu, and Renfei Zhou, *New Bounds for Matrix Multiplication: from Alpha to Omega*. SODA 2024; arXiv:2307.07970v2. `https://arxiv.org/abs/2307.07970v2`.

[14] OpenAI, *An Upper Bound of* $9/4$ *for the Matrix Multiplication Exponent*, October 2, 2026. Pinned at `openai/math@7ef5c0cc5c`. Paper source, including finite rank schemes and number-field remark.

[15] OpenAI, *Complex Matrix Multiplication Below 2.258 and Rectangular Bounds*, September 24, 2026. Same pinned revision. Paper and sources; Formalization scope.

[16] Anthropic, *A* $2^{o(n)}$ *Time Algorithm for 3-SAT*, pull request 31 at commit `1a6eee293b9e00cb432eef6018a21f523623a9af`. User-supplied changes; Paper and Lean project.

[17] Anthropic, `eth/MainResults.lean`, `eth/Solution.lean`, and checking instructions, at the same pinned commit.